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CARIA:
A Personalized Career Recommender Based on Individual Competency Similarity Measure

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ABSTRACT

The purpose of this research is to create a personalized system called CARIA that suggests career recommendations based on students' competencies and the required skills in each career. The focus of this study is on digital technology and digital media careers. The personalized career recommender system uses a novel similarity measure called modified Euclidean similarity to evaluate its performance and compare it with other similarity measures, machine learning, and GPT-4 techniques. The experimental results showed that modified Euclidean similarity achieved a precision@10 score of 0.83, which outperformed other techniques. The main objective of CARIA is to provide students with suitable career paths and conduct a competency gap analysis. This helps students choose a career path that fits their abilities. This research contributes to education in digital technology, digital media, and the workforce by providing employees with competencies that align with their needs.

KEYWORDS

Career Recommender, Digital Career, Individual Competency, Similarity Measures

INTRODUCTION

Digital careers play a vital role in developing and driving the country's economy. According to the 2019-2023 Thai working population survey conducted by the National Statistical Office of Thailand (2023), positions in digital careers increased from 115,000 to 161,000. In 2023, digital careers accounted for 1.60% of all 39.80 million workers in Thailand. Most individuals working in digital careers were between 25 and 49 years old, with 61.40% having graduated with a bachelor's degree or higher.

The future workforce will see a high demand for careers in the digital sector, many of which will offer job security and good financial rewards. According to a survey by LinkedIn (2024), the most sought-after careers will rely heavily on technology and digital skills, including roles such as

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artificial intelligence (AI) specialists and cyber-security experts. Similarly, a survey by Nexford University (2024) highlighted the top 10 highest-paying jobs, with digital careers like AI architect, data scientist, and software engineer manager being among the most prominent. These findings indicated a growing need for digital skills in businesses and industries, with employers strongly emphasizing pursuing digital careers.

Research studies and literature reviews have found that the rapid development of technology and the spread of COVID have led to a decline in employment in many countries. Additionally, the birth rate has decreased, and an aging society has led to a decline in youth entering the workforce. However, the workforce still requires people with digital skills or the ability to work effectively in a rapidly changing and competitive technology environment (José-García et al., 2023). Therefore, university students graduating to work in the future must carefully consider careers that suit their interests and match the workforce's needs.

Many students, especially those about to enter a university, need help choosing a career path (Ferry, 2006). When selecting a field of study and a career, some students consider societal and familial influences, personal aptitudes, and academic performance (Salami, 1999). Unfortunately, many students make career choices that do not align with their personality, interests, and abilities. They are often influenced by high school guidance counselors, leading to about 25% of university students experiencing financial and emotional stress. Additionally, approximately 75% of students end up changing their major at least once before graduation (Gordon, 2007). Research has indicated that career choices are complicated, and students require more information to make informed decisions. Each student has different abilities and skills, and they need to identify what skills they need to pursue a career (Sunil et al., 2023). Parents and relatives may offer advice, but they may need assistance guiding students toward new careers (Ochirbat et al., 2018; Rojsutee & Yongyuan, 2014; Talib & Aun, 2009). Often, students select a field or career based solely on the potential for a high income or growth opportunities, regardless of their abilities (Alimam et al., 2017). Career advice has been shown to help students identify their skills, abilities, interests, and values, leading to academic and career success (Martinez & Dănălache, 2008; Qamhieh et al., 2020; Vespia et al., 2018).

This paper introduces a personalized career recommender called CARIA, which uses individual competency similarity measures to provide tailored career recommendations. The paper's contributions include introducing a new similarity measure named modified Euclidean similarity (MES) for personalized career recommendations, developing a personalized career recommendation system called CARIA, and conducting extensive experiments to demonstrate the proposed MES's performance compared to other similarity measures, machine learning, and GPT-4 techniques.

LITERATURE REVIEW

Several studies have been conducted on career recommendation systems, which can be classified into four groups based on their recommendation techniques. The first group (Song et al., 2022; Sunil et al., 2023; Wanon et al., 2018) utilized data mining techniques for career recommendation. Sunil et al. (2023) proposed a methodology that uses machine learning techniques to consider the requirements of fresh graduates, including company demographic information, organization type, and job preferences. They utilized SVD (Singular Value Decomposition), SVD++ (an extension of SVD), and NMF (Non-negative Matrix Factorization) algorithms to create an automated recommendation system. This system assesses the candidate's background information, interest-based skills and abilities, and profile information to recommend appropriate jobs using mean absolute error (MAE) and root mean squared error (RMSE). Song et al. (2022) aimed to improve the accuracy of predicting career choices based on interest inventories using machine learning. They examined a large sample (N = 81,267) of employed and unemployed participants. They compared the accuracy of predicting occupational membership for the employed and vocational aspirations for the unemployed using a traditional interest profile method versus a new machine learning-augmented method. The results

indicated that the machine learning-augmented method performed better than the traditional profile method in predicting career choices. Wanon et al. (2018) studied career forecasting techniques for undergraduate students. There were nine input attributes, which were gender, father's career, father's income, mother's career, mother's income, average grades, field of study, special abilities, and correspondence between the career and the field of study. The system had three outputs of career groups: government officials, business owners, and business employees. Data analysis applied three techniques: decision tree, random forest (RF), and bagging. The results showed that RF gave the highest accuracy of 84.29%.

The second group, consisting of Cui (2024) and Qamhieh et al. (2020), used fuzzy logic to create their systems. Cui (2024) introduces a new career assessment and planning approach using the Sugeno fuzzy optimized weighted machine learning framework. The proposed framework combines fuzzy logic principles with machine learning techniques to offer tailored career recommendations based on individuals' skills, preferences, and values. By incorporating optimization techniques, the framework enhances the accuracy of the career assessment model, improving its ability to predict suitable career paths for individuals. Qamhieh et al. (2020) introduce a personalized career path recommendation system (PCRS) to assist high school students in selecting engineering disciplines. The PCRS is based on a fuzzy intelligence N-layered architecture, using students' academic performance, personality type, and extra-curricular skills. The evaluation results demonstrated a slight agreement between the recommendations provided by the PCRS and the actual disciplines chosen by the research sample.

The third group, consisting of Siddiky and Akter (2022) and Yidiz-Akyol and Boyaci (2020), utilized statistical techniques to recommend careers. Siddiky and Akter (2022) proposed a theory that explains students' career choices from a social-environmental perspective. Their study revealed that various factors, including family preferences, teachers' advice, job prestige, job security, remuneration, opportunities for promotion, pension benefits, professional development prospects, personal interests, academic majors, educational achievements, and career development training are significantly associated with students' career decision-making. Yidiz-Akyol and Boyaci (2020) investigated whether the future careers of university students can be significantly predicted by their cognitive flexibility and positivity. The analyses were conducted to ascertain the predictability of students' career futures based on cognitive flexibility and positivity using an analysis of variance (ANOVA). The results revealed that cognitive flexibility and positivity significantly predict university students' future careers.

In another group of studies (Gugnani & Misra, 2020; José-García et al., 2023; Mankawade et al., 2023; Ochirbat et al., 2018), similarity measures were used to develop career recommendation systems. Similarity measures are statistical techniques used to measure the similarity between two objects. José-García et al. (2023) introduced C3-IoC, an AI-based solution designed to help students explore information technology career paths based on their education level, skills, and prior experience. This tool presents a new similarity metric method for connecting existing job roles to technical and non-technical skills. Additionally, C3-IoC allows users to visualize a network of job roles and find their place within communities of job roles. Mankawade et al. (2023) introduced an architecture designed to identify the most suitable professions based on an individual's resume. They utilized natural language processing and machine learning to train a model that could predict the profession based on the resume. Subsequently, jobs were scraped from Naukri.com using the predicted profession as the search keyword. The required skills for these jobs were then compared to the individual's skills using a cosine similarity algorithm. The matched skills were ranked and presented to the user. The accuracy of the proposed model is 97%. Gugnani and Misra (2020) proposed a job matching system that can pair resumes with non-standard or unstructured/semi-structured job descriptions (JDs). The system uses natural language processing techniques for skill extraction and introduces the concept of extracting implicit skills. These implicit skills are not explicitly mentioned in a JD but may be implied based on geography, industry, or role. The system identifies other JDs like the given one and performs a similarity match based on a semantic space. The research explores various similarity

measures to match the skills extracted from a candidate's resume to the explicit and implicit skills of JDs. Empirical results show that this approach improves matching by 29.4% compared to a baseline method that uses only explicit skills. Ochirbat et al. (2018) developed a career recommendation system that used demographic data (such as name, year of birth, gender, country, school of study, field of expertise, and interests) and data collected from a questionnaire that summarized the scores into personality types. The researchers used five similarity measures, including Euclidean, intersection, cosine, Jaccard, and Pearson, to calculate scores. According to their results, Pearson had the highest user satisfaction, followed by Euclidean.

In the domain of career recommendation systems, several techniques have been employed to categorize them. Research related to these systems has been compared in Table 1, based on their target audience, input, output, and evaluation methods.

Most of the recommenders are designed for undergraduate students, taking into consideration their demographic information and personality traits as input. While some can recommend only career groups, a few can identify appropriate specific careers for students. Data mining techniques and similarity measures are commonly used in these models.

This research aims to develop a personalized career recommender, named CARIA, by applying a novel similarity measure. The recommender ranks digital careers for individuals and analyzes the competency gap between individuals and these careers. This article also compares the proposed technique with other similarity measures, machine learning, and GPT-4 techniques for digital career recommendation performance.

CARIA SYSTEM FRAMEWORK

This research project aims to develop a personalized career recommendation system called CARIA, which consists of five key processes as illustrated in Figure 1. The first process involves creating a questionnaire to gather information about an individual's competencies and career preferences. The second process involves collecting personal and career-related data from digital career experts. The third process involves developing a personalized career recommendation system, which is tailored to the individual's competencies and preferences. The fourth process evaluates the system's performance using various similarity measures, machine learning algorithms, and GPT-4 techniques. Finally, the fifth process involves applying the recommendation system to students. The system begins by asking students to complete a psychological questionnaire to evaluate their competencies. The system then compares the student's competencies with those required for each career and recommends suitable options. In addition to ranking the top 10 appropriate careers, the system also analyzes the gap in the student's competencies and those needed for the recommended careers.

Creating a Questionnaire to Collect Competency Data

This study is aimed at collecting data on the competence of individuals who are pursuing digital careers. The data was collected across three domains: skills, knowledge, and attitudes. To do this, 35 psychological questions from the MyMajors (2022) website and 33 questions from the Department of Labor website (USA O*NET Development, 2022) were gathered and compiled into questionnaires. These questionnaires were evaluated by five experts to ensure their suitability for assessing the competence of individuals. The item objective congruence index was used to verify the effectiveness of each question (Nielsen & Landauer, 1993), and each question needed an item objective congruence index of 0.5 or above. After experts' evaluation, the questionnaire consisted of 26 questions about skills and attitudes and 29 questions about knowledge. Further, it was suggested that the 26 skill and attitude questions that used the same question could not determine to measures skill or attitude.

The assessment consisted of 26 questions, which were divided into two parts, namely skill and attitude sections. Both sections contained 26 questions each. The skill section questions were structured with a "can" statement, such as "Can you help others solve problems?" In contrast, the

Table 1. A comparison of research related to the career recommendation system

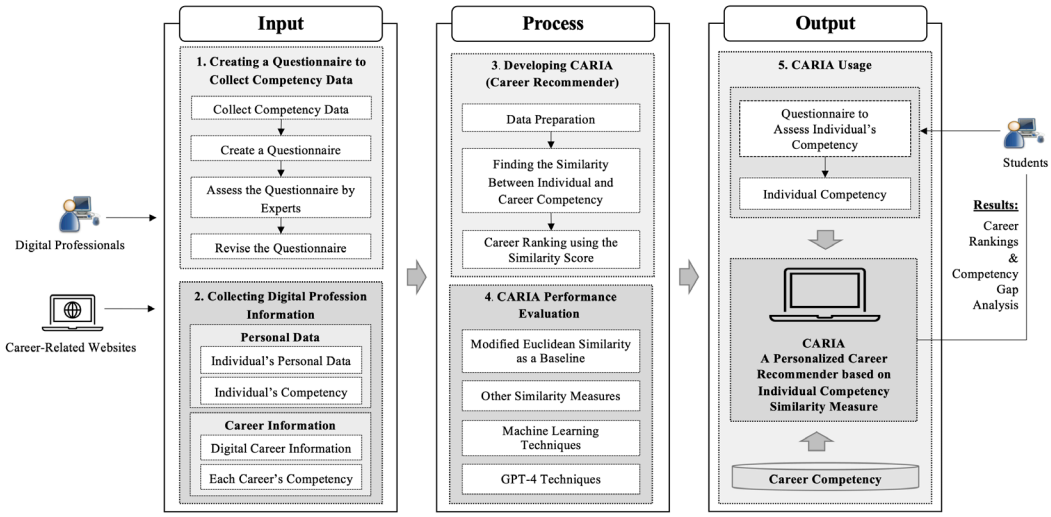
Career recommendation	Related work											CARIA
	Cui (2024)	José-García et al. (2023)	Mankawade, 2023)	Sunil et al. (2023)	Siddiky & Akter (2022)	Song et al. (2022)	Gugnani & Misra (2020)	Qamhieh et al. (2020)	Yidiz-Akyol & Boyaci (2020)	Ochirbat et al. (2018)	Wanon et al. (2018)	
Target group												
High school students								✓				
Undergraduate students	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓
Postgraduate students		✓				✓						
Input												
Demographics/Profile	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Employment status						✓					✓	
Knowledge		✓	✓					✓		✓		✓
Skill	✓	✓	✓	✓			✓	✓		✓		✓
Attitude/Personality	✓	✓	✓					✓		✓		✓
Techniques												
Classification				✓							✓	✓
Clustering						✓						
Fuzzy logic	✓							✓				
ANOVA									✓			
Chi-square					✓							
Similarity measures		✓	✓				✓			✓		✓
Output												
Career group	✓				✓	✓		✓	✓		✓	
Specific career		✓	✓	✓			✓			✓		✓
Evaluation methods												
Accuracy	✓	✓	✓								✓	

continued on following page

Table 1. Continued

Career recommendation	Related work											CARIA
	Cui (2024)	José-García et al. (2023)	Mankawade, 2023)	Sunil et al. (2023)	Siddiky & Akter (2022)	Song et al. (2022)	Gugnani & Misra (2020)	Qamhieh et al. (2020)	Yidiz-Akyol & Boyaci (2020)	Ochirbat et al. (2018)	Wanon et al. (2018)	
Precision@K						✓	✓			✓		✓
Support and confidence												
MAE				✓						✓	✓	
- RMSE				✓						✓	✓	
Satisfaction/Usability testing		✓						✓		✓		✓

Figure 1. A framework for a personalized career recommendation system



attitude section questions contained a "like" statement—for example, "Do you like to help others solve problems?" The questions were reviewed and evaluated by experts, and the assessment results found them appropriate.

The online questionnaire consisted of 81 questions, which were further classified into 26 skill, 26 attitude, and 29 knowledge questions. The questionnaire was structured into three parts. The first part contained five personal information question categories, such as gender, current career, period of career, cumulative grade point average, and field of study. The second part consisted of 81 questions to assess individual competency data, and the respondents were required to give a score from zero to 100, with zero being the lowest level of competency and 100 the highest. The third part consisted of 10 questions about career information of interest. The assessment used 96 questions from all three parts to assess students' competencies.

Collecting Digital Profession Information

This step bifurcated the data collection process into two categories, personal data and career information, which were then used to design a personalized career recommendation system. The recommendation system is explained below.

Personal Data Collection

This study gathered personal data from 100 experts, divided into two groups: 50 experts in digital technology (DT) careers and 50 in digital media careers. Informed consent was obtained from all of the subjects involved in the study. Of the 100 experts, 36 were males and 64 were females. Of these experts, 13 had worked for less than a year, 30 for one to three years, 35 for four to six years, 12 for seven to nine years, and 10 for more than 10 years. The data was collected using an online questionnaire between January 2023 and June 2023. The survey responses were validated to ensure that the online questionnaire accurately reflected the intended measures. Reliability checks were conducted to ensure consistent results for repeated applications under similar conditions. In this research, the method for assessing the validity and reliability of survey responses involved conducting re-interviews with respondents at a short interval (one to two weeks) after an initial interview and estimating relatively simple statistics from the data. The correlation between responses to the same question asked at two different time points was used to estimate reliability (Tourangeau, 2020).

Table 2. Sample questions assessed by an expert with one respondenty a competency scores

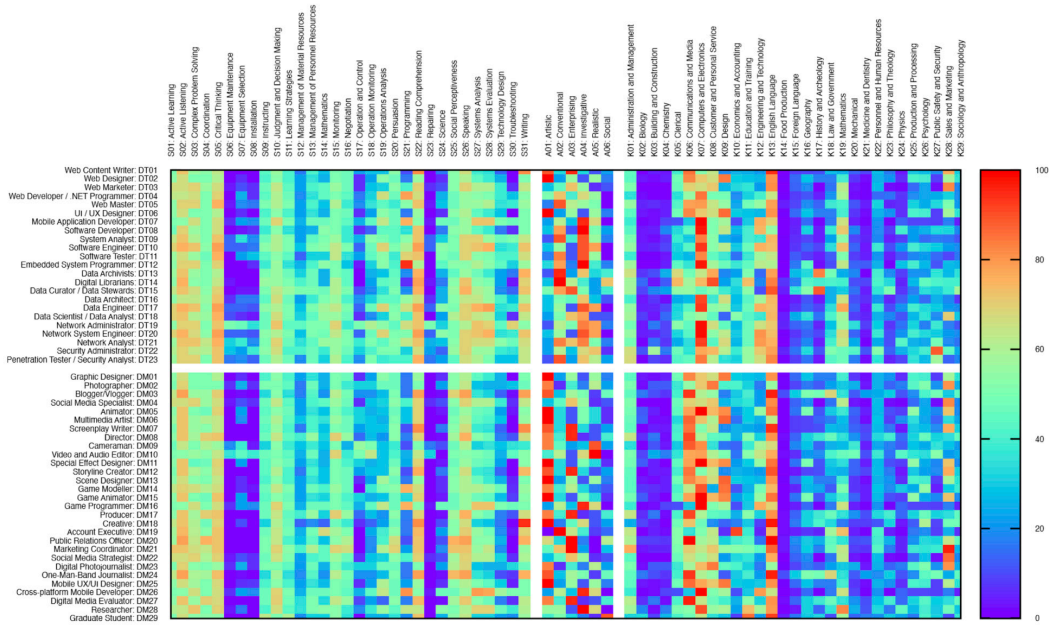
Questions	Score (0-100)	Competency	Average score
Skill			
You can learn by yourself	92	Active learning	92.00
You can work in a team. You can provide nursing care.	83 96	Active listening	89.50
…	…	…	…
You can write a narrative from your imagination. You can write a narrative from your experiences.	87 14	Writing	50.50
Attitude			
You like to write a narrative from your imagination. You like to write a narrative from your experiences. You like to rely on your design thinking.	21 72 80	Artistic	57.67
You like to operate a machine. You want to combine work with learning. You like to learn, work hard, and have discipline. You like to work with a detailed, logical approach.	15 72 29 17	Conventional	33.25
…	…	…	…
You enjoy working with organizations and groups. You like to work in a team. You enjoy conveying information to others. You enjoy being involved in current events.	76 11 77 95	Social	64.75
Knowledge			
You understand administration and management.	38	Administration	38.00
You understand biology.	97	Biology	97.00
…	…	…	…
You understand sociology and anthropology.	82	Sociology	82.00

The online questionnaire was divided into three parts. The first part collected personal data from respondents, including their gender, current career, period of career, cumulative grade point average, and field of study. The second part, as shown in Table 2, contained information on individual competency obtained by respondents answering questions about competency in three areas: skill, knowledge, and attitude. The third part assessed the career suitability suggested by the model. The first and second parts were used to construct the career recommendation system, and the third was used to assess the system.

Career Information Collection

This dataset was split into two parts. The first part consisted of data on the necessary competencies for DT and digital media careers, as specified on the website of the Department of Labor. These competencies included 35 skills, six attitudes, and 33 knowledge areas. Each competency was scored between zero to 100, with a score of 50 or higher indicating its essentiality for the career. We employed these competencies to develop an online questionnaire, which was evaluated by five experts to determine the appropriate competencies for each career. After analyzing the evaluation results, we concluded that for each career, there were 31 skills, six attitudes, and 29 knowledge areas that were suitable, as shown in Figure 2. Some skills and knowledge areas were eliminated because the experts deemed them irrelevant to the digital careers.

Figure 2. Competency of digital careers



The second part of the collection provided information about the career outcomes of DT and digital media curricula offered at Suranaree University of Technology in Thailand. We have categorized DT and digital media careers into six and nine groups, respectively, based on their similar competencies. This categorization has resulted in a total of 36 careers in DT and 42 careers in digital media, which are presented in Table 4. Furthermore, Table 3 provides an example of career competency scores for skill, attitude, and knowledge.

Developing a Personalized Career Recommendation System

The personalized career recommendation system was created using the Java web framework and MySQL database. It is currently hosted on a machine equipped with an Intel Core i7 processor running at 2.7 GHz and 16 GB of memory. This study utilized a hybrid agile-waterfall methodology for software development (SD) life cycles (Reiff & Schlegel, 2022). The hybrid methodology integrates the waterfall and agile models to follow a sequential flow for requirements and design, with iterative cycles, known as sprints, for the implementation process. A sprint is a time-boxed execution in which teams develop and implement product features of the CARIA recommender. Eventually, the waterfall model validates and maintains the recommender. Therefore, the hybrid methodology combines the precision of the waterfall model and the speed of the agile model, making it the best of SD models. Figure 3 illustrates the process of the hybrid agile-waterfall methodology used to develop the CARIA recommender.

The developed system utilizes a method to measure the resemblance between the competencies of students and careers in order to provide personalized recommendations. This resemblance is calculated by determining the distance between the competency scores of students and each career. If the distance between two data points is small, it indicates a high degree of similarity between the objects, and if the distance is large, it indicates a low degree of similarity. A high similarity score between a student's competencies and a career suggests that the student should consider pursuing that career. The following steps were taken to create the digital career recommendation system.

Figure 3. Hybrid agile-waterfall methodology for developing the CARIA recommender

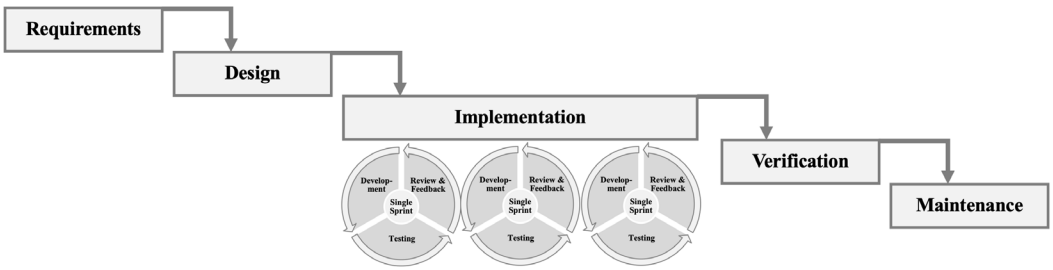


Table 3. An example of career competency scores

Identity	Competency	Score (0-100)		
		Web content writer	Web designer	Web marketer
Skill				
1	Active learning	50.00	53.00	66.00
2	Active listening	72.00	66.00	66.00
...	...	...	...	...
31	Writing	81.00	50.00	63.00
Attitude				
1	Artistic	89.00	100.00	28.00
2	Conventional	33.00	22.00	45.00
...	...	...	...	...
6	Social	22.00	17.00	11.00
Knowledge				
1	Administration	45.00	44.00	51.00
2	Biology	0.00	3.00	0.00
...	...	...	...	...
29	Sociology	31.00	36.00	18.00

For Step 1, in order to prepare the data, we need to compare the individual competency scores, which are listed in Table 2, with the career competency scores in Table 3. Based on this comparison, we will then adjust the student competency score using Eq. (1):

$$C'_{stu} = \begin{cases} C_{car}, & if\, C_{stu} \geq C_{car} \\ C_{stu}, & otherwise \end{cases} \tag{1}$$

If a student competency score (C_{stu}) is greater than or equal to the career competency score (C_{car}), the student competency score will be adjusted to equal the career competency score. The same score means the student is already competent to pursue that career. However, no score adjustments are required if a student competency score (C_{stu}) is less than the career competency score (C_{car}). Then, the adjusted student competency scores (C'_{stu_i}) will be used to measure distance in the next step.

Table 4. Examples of career groups and careers in digital technology and digital media

Career group	Career
DT	
Web application	Web content writer, web designer, web marketer…
Enterprise software	Software developer, system analyst, software engineer…
...	...
Cloud technology	Network administrator, penetration tester, security analyst…
Digital media	
Digital content	Digital copy writer, blogger, Youtuber…
Animation	Animator, modeler, character designer…
...	...
Digital media research	Digital media evaluator, researcher, graduate student…

For Step 2, using the adjusted student competency scores (C'_{stu}) from Table 5, we calculate similarity scores between student and career competencies using a similarity measure called MES, as formulated in Eq. (2):

$$\text{MES}\,(C'_{stu},\ C_{car}) = \frac{1}{1 + \sqrt{\sum_{i=1}^{n}\left(C'_{stu_i} - C_{car_i}\right)^2}} \tag{2}$$

where C'_{stu_i} denotes the adjusted student competency score in each factor i, C_{car_i} denotes the career competency score in each factor i, and n denotes the total number of competencies ($n = 66$ in this research).

In this example, we will discuss how to apply the MES technique using Eq. (2), as shown in Figure 4. The figure illustrates the process of calculating the similarity scores between a student and various career options. In practice, this process involves assessing 66 competencies, which include 31 skills, six attitudes, and 29 knowledge areas.

Step 3 involves ranking the careers based on the similarity scores calculated in Step 2. If a student's similarity score with a particular career is small, the career suits the student. The most appropriate careers for the student are then listed in ascending order of similarity score, as shown in Table 6. Additionally, the results are presented as a career recommendation graph, as shown in Figure 5.

Furthermore, the system can also analyze the gap between student and career competencies. This analysis compares the details of each career to see how much students lack competencies, as demonstrated in Figure 6.

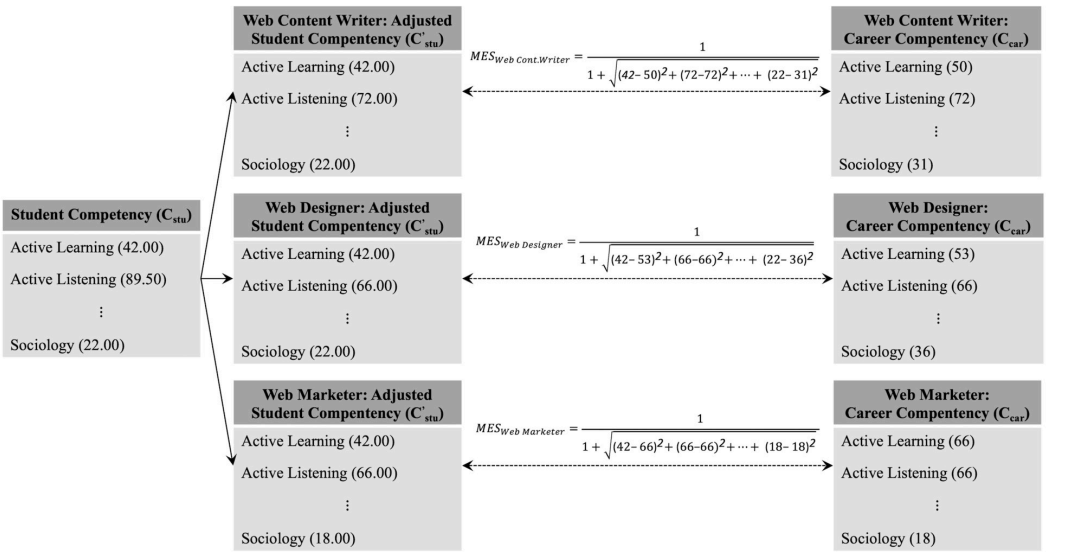
EXPERIMENTAL RESULTS AND DISCUSSION

The CARIA performance is assessed based on two aspects: the evaluation of the recommendation technique and the evaluation of the recommendation system's usability. The experimental results are discussed below.

Recommendation Technique Evaluation

The CARIA recommender utilizes a new similarity measure called MES to assess its performance and compare it with other similarity measures, machine learning, and GPT-4 techniques. These techniques were assessed using an evaluation metric called precision@K. The precision@K is a

Figure 4. An example of calculating the similarity scores between a student and each career

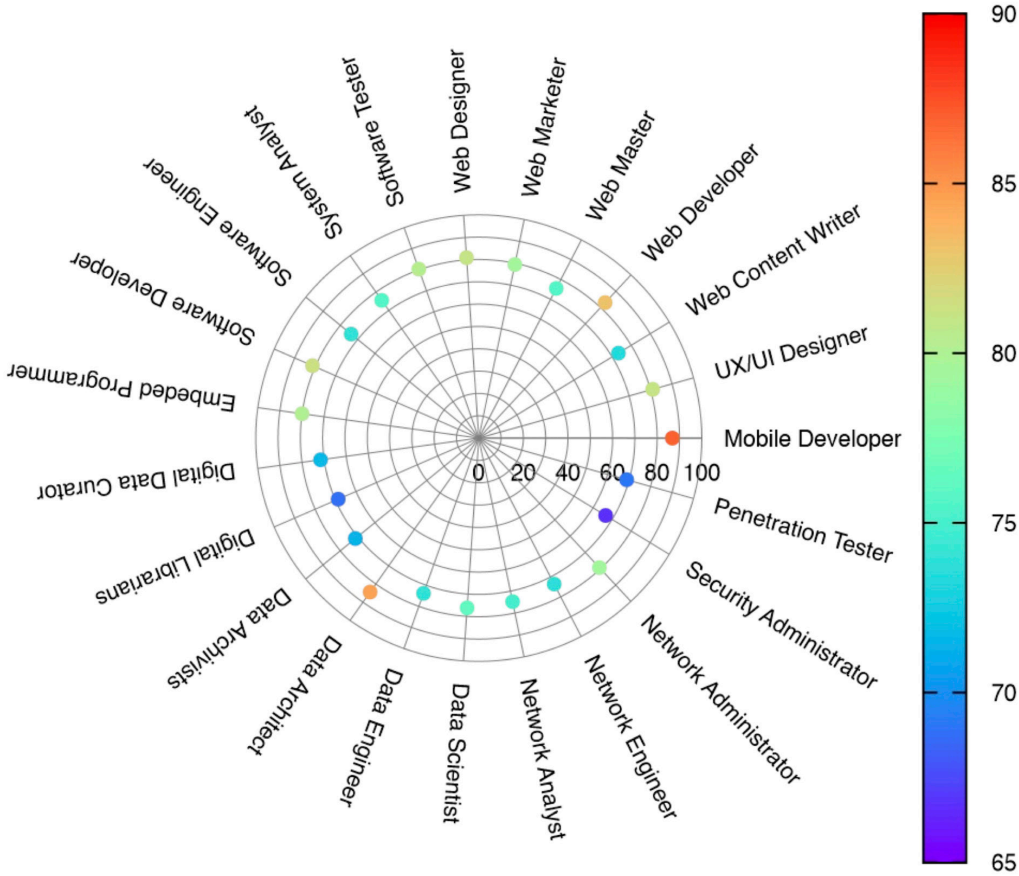


standard evaluation metric used in recommendation systems to measure the quality of ranked results (Tamm et al., 2021). It quantifies the proportion of relevant recommended items in a recommendation list of size K. It is formulated in Eq. (3):

Table 5. Examples of adjusted student competency scores

Identity	Competency	Student competency score C_{stu}	Score (0-100)					
			Web content writer		Web designer		Web marketer	
			C_{car}	C'_{stu}	C_{car}	C'_{stu}	C_{car}	C'_{stu}
Skill								
1	Active learning	42.00	50.00	42.00	53.00	42.00	66.00	42.00
2	Active listening	89.50	72.00	72.00	66.00	66.00	66.00	66.00
...	...	...	...	...	...	...	...	...
31	Writing	50.50	81.00	50.50	50.00	50.50	63.00	50.50
Attitude								
1	Artistic	57.67	89.00	57.67	100.00	57.67	28.00	28.00
2	Conventional	33.25	33.00	33.00	22.00	22.00	45.00	33.25
...	...	...	...	...	...	...	...	...
6	Social	64.75	22.00	22.00	17.00	17.00	11.00	11.00
Knowledge								
1	Administration	38.00	45.00	38.00	44.00	38.00	51.00	38.00
2	Biology	97.00	0.00	0.00	3.00	3.00	0.00	0.00
...	...	...	...	...	...	...	...	...
29	Sociology	22.00	31.00	22.00	36.00	22.00	18.00	18.00

Figure 5. An example of a career recommendation graph



$$Precision@K = \frac{\text{Number of relevant items in } K}{\text{Total number of items in } K} \quad (3)$$

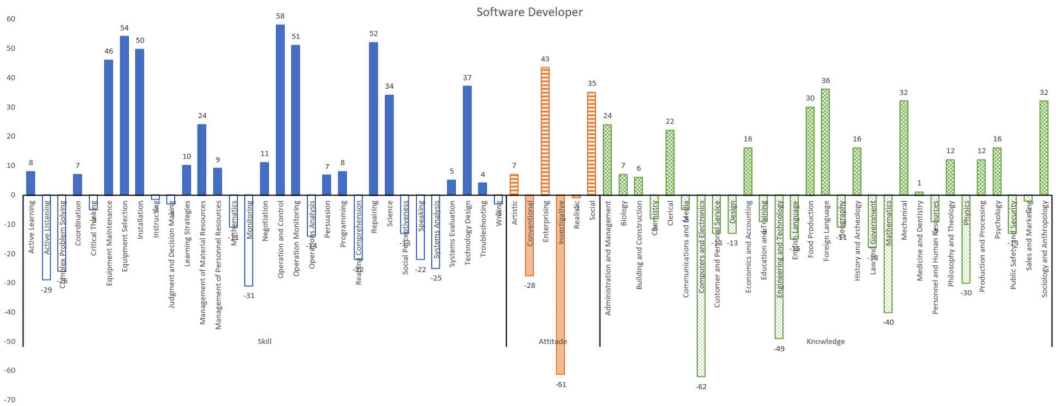
The precision@10 evaluation is performed when the system recommends the top 10 careers. The actual test cases are shown in Table 7. The CARIA metric was evaluated with two datasets, one for DT careers and another for digital media careers, to ensure unbiased and over-fitting results.

Table 8 demonstrates an example of an evaluation case where a checkmark (✓) in the CARIA column indicates that the predicted results of CARIA are in the top 10 of the actual test case. Thus, the precision@10 of this example is 6/10 = 0.6.

Table 9 reveals that the evaluation result of the DT career was 0.89 at precision@10, and the digital media career had a precision of 0.76 at precision@10. The precision@10 value averaged from 100 digital careers was 0.83.

After that, the career recommendation system was improved by searching for the best techniques. The MES technique was used as a performance comparison baseline since most previous research works have relied on it. The proposed MES technique was compared with other similarity measures, machine learning, and GPT-4 techniques.

Figure 6. An example of gap analysis between student and career competencies



Performance Comparisons Results With Other Similarity Measures

A research study compared different similarity measures with the MES technique. These measures included Jaccard similarity, cosine similarity, and Pearson correlation techniques (Baisantry & Shukla, 2017; Saranya et al., 2016). The Jaccard similarity is used to compare document data in text mining. Its value ranges from zero to one, where zero indicates identical documents while one means no

Table 6. An example of career rankings using the MES technique

Career	Similarity score	
	Score	Rank
DT07: Mobile developer	0.231	1
DT16: Data architect	0.219	2
DT04: Web developer/.NET programmer	0.213	3
DT08: Software developer	0.205	4
DT02: Web designer	0.204	5
DT06: UI/UX designer	0.204	6
DT11: Software tester	0.201	7
DT12: Embedded system programmer	0.200	8
DT03: Web marketer	0.198	9
DT19: Network administrator	0.197	10
DT18: Data scientist/Data analyst	0.186	11
DT09: System analyst	0.183	12
DT05: Web master	0.183	13
DT21: Network analyst	0.181	14
DT10: Software engineer	0.178	15
DT17: Data engineer	0.178	16
DT20: Network system engineer	0.177	17
DT01: Web content writer	0.176	18

Table 7. Examples of actual test cases

Test case identity	Skill				Attitude				Knowledge				Top 10 careers
	S01	S02	···	S31	A01	A02	···	A06	K01	K02	···	K29	
Test case in digital technology careers													
1	48	63	···	74	67	55	···	43	65	30	···	62	DT02, DT04, DT05, DT01, DT06, DT07, DT11, DT23, DT15, DT09
2	100	87	···	80	77	75	···	70	83	78	···	62	DT08, DT10, DT12, DT07, DT04, DT05, DT02, DT22, DT20, DT21
3	15	93	···	51	58	33	···	67	38	82	···	82	DT19, DT22, DT20, DT13, DT15, DT14, DT18, DT09, DT16, DT17
···	···	···	···	···	···	···	···	···	···	···	···	···	···
50	100	100	···	90	80	91	···	77	71	80	···	85	DT11, DT21, DT22, DT23, DT20, DT13, DT09, DT15, DT05, DT18
Test case in digital media careers													
1	71	74	···	72	75	78	···	72	78	82	···	92	DM04, DM09, DM19, DM26, DM02, DM03, DM22, DM17, DM06, DM11
2	95	75	···	74	77	72	···	44	95	99	···	34	DM01, DM08, DM12, DM03, DM05, DM20, DM17, DM04, DM11, DM18
3	97	93	···	71	78	94	···	86	81	67	···	78	DM05, DM02, DM26, DM08, DM12, DM11, DM24, DM01, DM07, DM18
···	···	···	···	···	···	···	···	···	···	···	···	···	···
50	63	72	…	50	58	61	…	64	70	36	…	43	DM24, DM11, DM03, DM01, DM27, DM23, DM06, DM14, DM09, DM22

Table 8. An example of an evaluation case

Top 10 rank	Test case	CARIA
1	DT16: Data architect	**✓DT02: Web designer**
2	**DT07: Web developer**	**✓DT07: Web developer**
3	**DT02: Web designer**	**✓DT05: Web master**
4	**DT06: UX/UI designer**	DT01: Web content writer
5	**DT11: Software tester**	**✓DT06: UX/UI designer**
6	DT08: Software developer	**✓DT07: Mobile developer**
7	**DT05: Web master**	**✓DT11: Software tester**
8	DT03: Web marketer	DT23: Penetration tester
9	**DT07: Mobile developer**	DT15: Digital data curator
10	DT18: Data scientist	DT09: System analyst

Table 9. The evaluation results of a personalized career recommendation system

Digital career	Precision@10
DT	0.89
Digital media	0.76
Average	0.83

similarity. On the other hand, the cosine similarity is used to measure the similarity of non-binary vectors and ignores zero-zero matches like the Jaccard measure. The Pearson correlation, also known as Pearson similarity, is the cosine similarity of two vectors after subtracting the means of each.

This research modified three similarity measures to calculate the similarity between the competencies of each student and each career. These measures are the modified Jaccard similarity, modified cosine similarity, and modified Pearson similarity techniques, as formulated in Eq. (4), Eq. (5), and Eq. (6):

$$\text{MJS}\ (C'_{stu},\ C_{car}) = \frac{\sum_{i=1}^{n} C'_{stu_i} C_{car_i}}{\sum_{i=1}^{n}\left(C'_{stu_i}\right)^2 + \sum_{i=1}^{n}\left(C_{car_i}\right)^2 - \sum_{i=1}^{n} C'_{stu_i} C_{car_i}} \tag{4}$$

$$\text{MCS}\ (C'_{stu},\ C_{car}) = \frac{\sum_{i=1}^{n} C'_{stu_i} C_{car_i}}{\sqrt{\sum_{i=1}^{n}\left(C'_{stu_i}\right)^2} \times \sqrt{\sum_{i=1}^{n}\left(C_{car_i}\right)^2}} \tag{5}$$

$$\text{MPS}\ (C'_{stu},\ C_{car}) = \frac{\sum_{i=1}^{n}\left(C'_{stu_i} - \overline{C'_{stu}}\right)\left(C_{car_i} - \overline{C_{car}}\right)}{\sqrt{\sum_{i=1}^{n}\left(C'_{stu_i} - \overline{C'_{stu}}\right)^2\left(C_{car_i} - \overline{C_{car}}\right)^2}} \tag{6}$$

where C'_{stu_i}, C_{car_i} and n are the exact definition of Eq. (2), $\overline{C'_{stu}}$ *and* $\overline{C_{car}}$ are the average values of C'_{stu_i} and C_{car_i}, respectively.

The system rates the careers suitable for each student using various techniques and evaluates them through digital career test cases. The similarity scores of the different techniques are compared with the MES technique, which is considered as the baseline. The performance of other similarity measures (Δ_{other}) is evaluated using the precision@10 score as Eq. (7):

$$\Delta_{other} = \frac{Precision@10_{other} - Precision@10_{MES}}{Precision@10_{MES}} \times 100 \tag{7}$$

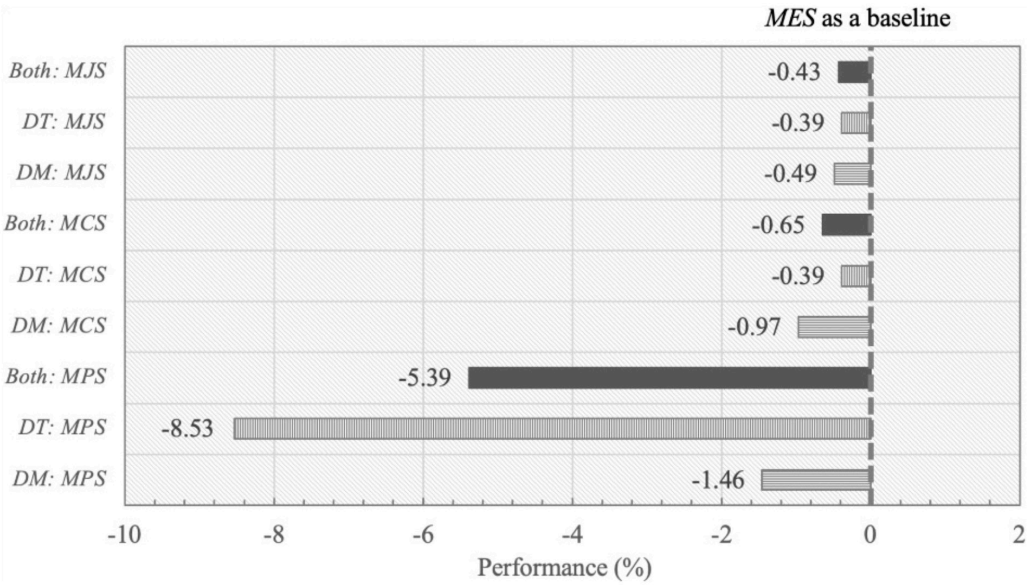
The results, as shown in Figure 7, indicate that the MES technique performs better than other techniques for career recommendations. The MES technique is an improvement from the MED technique and is advantageous because it provides accurate results for career recommendations. Therefore, the MES technique is the most suitable method for recommending careers.

Performance Comparisons Results With Machine Learning Techniques

Machine learning is often used to classify data by training a model to predict the category or class of a given input data point based on its features. This approach can also be applied in recommendation systems, where the predicted class is the output, precisely the top one ranking (Najafabadi et al., 2019). In this study, the decision trees (J48), naïve Bayes, and multilayer perceptron were compared to the MES technique. The training set consisted of all competencies, including 31 skills, six attitudes, and 29 knowledge, while the output variables were each test case's digital career, as shown in Table 7. However, in this case, the output class was top one, which is the career with the highest recommended score. The model was constructed using WEKA (Waikato Environment for Knowledge Analysis), an open-source application for data analysis (Ibrahim & Shiba, 2019).

The same test set was used as the training set to assess the model, and the 10-fold cross-validation approach was applied. The model's performance was compared to the MES technique using the

Figure 7. Performance of other similarity measurement techniques compared with the MES technique



precision@1 score. This score represents the output of the machine learning technique, which is a digital career, as shown in Eq. (8):

$$\Delta_{ML} = \frac{Precision@1_{ML} - Precision@1_{MES}}{Precision@1_{MES}} \times 100 \tag{8}$$

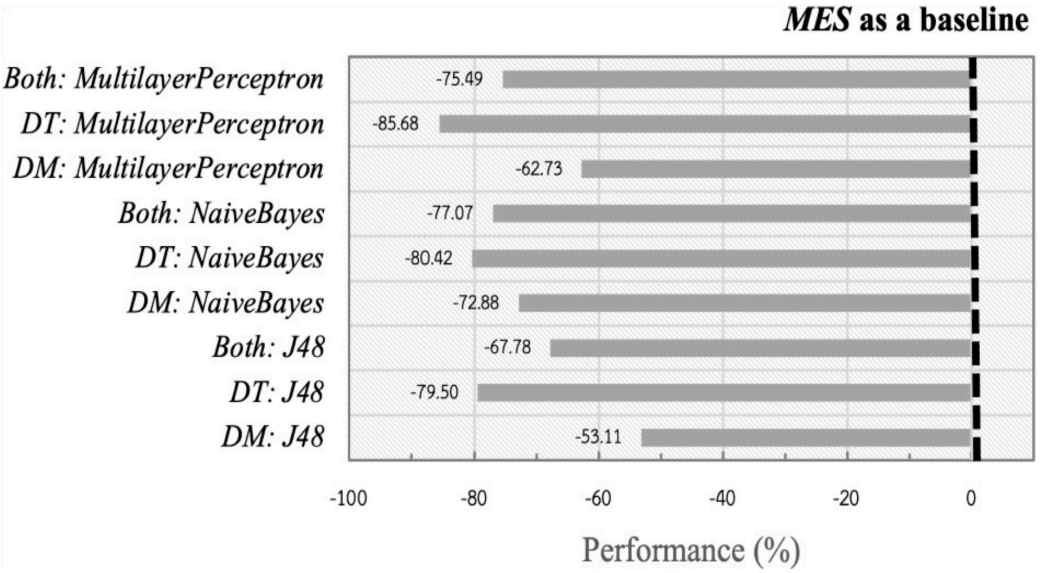
Figure 8 shows a comparison of the performance of machine learning techniques with the MES technique for career recommendations. The results indicate that the MES technique outperforms all machine learning techniques.

Performance Comparison Results With GPT-4 Techniques

Large language models such as GPT-4 have recently been introduced, and they have revolutionized various systems including recommendation systems. According to recent studies by Li et al. (2023), Bao et al. (2023), and Trichopoulos et al. (2023), the performance of the MES recommendation of CARIA has been compared with the recommendations generated by GPT-4. GPT-4 uses three recommendation techniques: k-nearest neighbor (KNN), neural network (NN), and RF. KNN is typically used for classification or regression. In order to adapt KNN for recommendation purposes, GPT-4 treats each career as a unique label associated with its scores. The model is then trained using the career data as input, with each career's scores serving as features. The top 10 career matches for each student are subsequently determined by the KNN model.

In adapting KNN for recommendation purposes, GPT-4 treats each career as a unique label associated with its scores. The model is then trained using the career data as input, with each career's scores serving as features. Finally, the top 10 career matches for each student are determined by the KNN model. Similarly, NN is also inherently a classification or regression model. GPT-4 adapts NN to perform career ranking by learning to predict probabilities or scores that reflect the relative ranking of careers. Hence, the network's output can be interpreted as a career ranking. The GPT-4 also uses RF for career recommendations. However, it employs a different approach by treating the career recommendation problem as a classification issue with a twist. Instead of merely classifying

Figure 8. Performance of the machine learning techniques compared with the MES technique



each student into a single career, it uses the model's probability estimates for each class (career) as a basis for ranking. The idea behind this is to train a RF classifier on the career data, predicting the likelihood of each career for a given set of scores, and then rank these careers based on the predicted probabilities for each student.

The model construction training set includes input variables comprising competencies such as 31 skills, six attitudes, and 29 knowledge as well as output variables corresponding to each digital career, as shown in Table 3. ChatGPT plus is employed to construct the GPT-4 model. The precision@10 score is used to evaluate the performance of GPT-4 compared to the MES technique. As depicted in Figure 9, the results indicate that the MES technique outperforms all GPT-4 generated techniques for career recommendations.

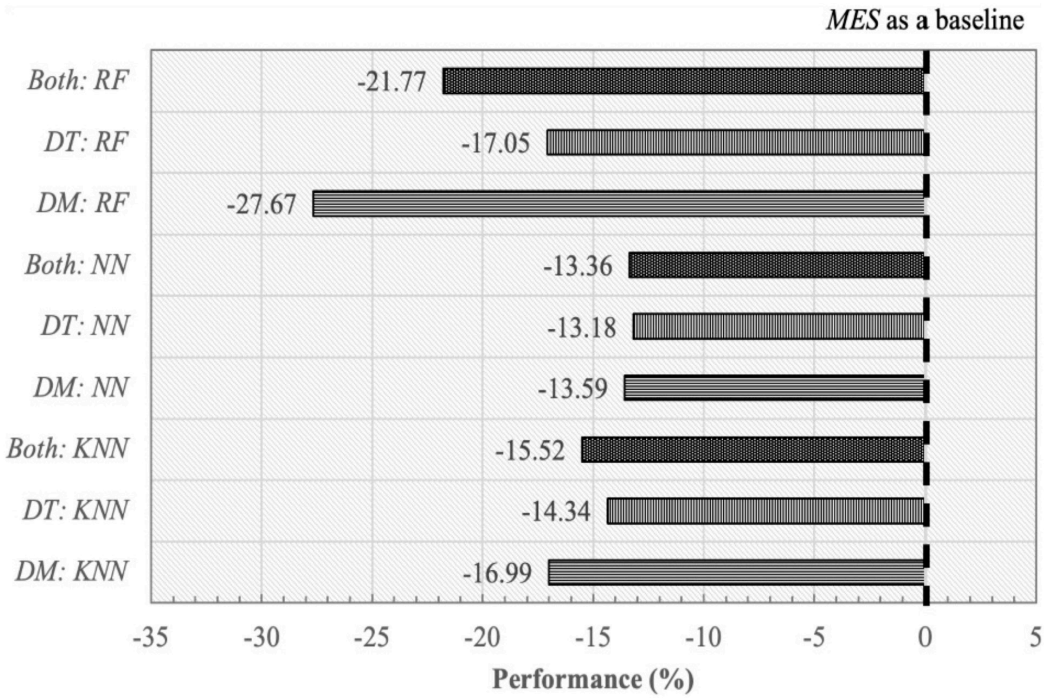
Recommendation System Usability Evaluation

The usability of the CARIA recommender was evaluated using the software usability measurement inventory (SUMI) (Kirakowski & Corbett, 2006). The SUMI assesses system usability in five aspects: affect, control, effectiveness, helpfulness, and learnability. Each aspect includes 10 questions, totaling 50, and utilizes a three-level scale (1-3). There are 20 positive questions (agree = 3, not sure = 2, disagree = 1) and 30 negative questions (agree = 1, not sure = 2, disagree = 3). The assessment results were then categorized into three levels: good (2.34– 3.00), fair (1.67–2.33), and poor (1.00–1.66), using a rating threshold of 0.66.

In a study by Nielsen and Landauer (1993), it was found that nine test users can uncover 95% of usability problems. The CARIA recommender underwent evaluation by nine experts, with three specializing in DT, three in digital communication, and three in SD. All of the experts had at least five years of experience in their respective fields. The evaluation was conducted using a laptop with a 14-inch screen and a web browser, using the 50 questions of SUMI tools.

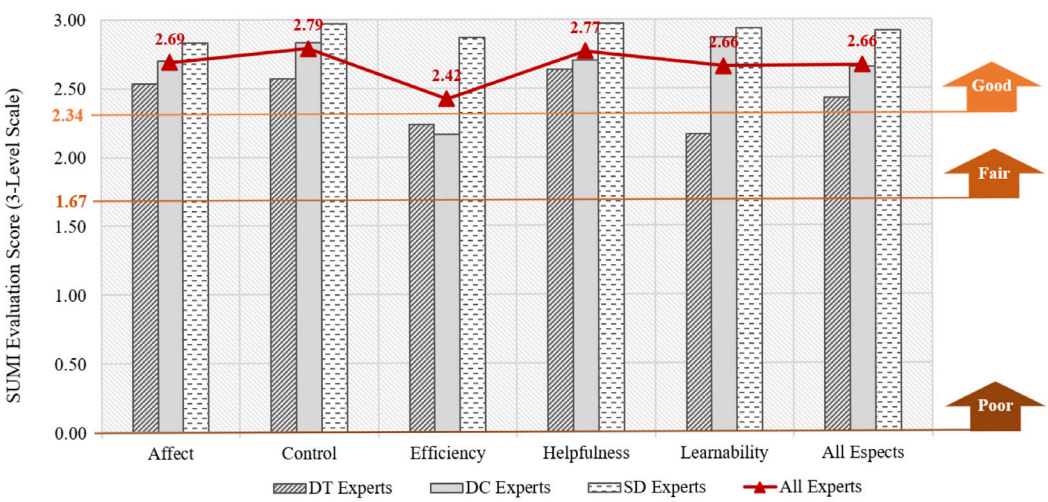
The assessment results show that the system's usability is generally good, with an average score of 2.66 and a standard deviation of 0.56, as displayed in Figure 10. A small standard deviation indicates consistent user responses. The system's control achieved the highest score among the five areas, with an average score of 2.79 and a standard deviation of 0.46. However, the DT group assessed the

Figure 9. Performance of the GPT-4 generated techniques compared with the MES technique



system lower than the other groups, particularly in terms of efficiency and learning ability. Further interviews uncovered the need to improve system effectiveness by reducing the number of steps required to perform tasks. The learning ability could also be improved by reducing the amount of information that users need to read before using the system.

Figure 10. SUMI evaluation results of CARIA recommender



DISCUSSION

Using similarity measures for career recommendation systems is more suitable than machine learning when the desired result is a top k rank (where $k > 1$). This is because machine learning is limited to the top one rank due to the nature of classification. Additionally, machine learning requires a large amount of data to create a model, which can be challenging. The data used in this research was collected from digital experts in the field.

The GPT-4 algorithm has adapted the classification to make top-k ranking possible. However, similarity techniques designed for ranking are better suited for this purpose than GPT-4. The proposed MES career recommendation model, which measures the similarity between students' competencies and the competencies needed for each career, is the best option for the CARIA recommendation system.

An analysis of the results reveals that there may be some issues with the competency requirements for each career in different countries. The competency of digital careers used in MES of CARIA is based on data from MyMajors, which is specific to the United States. However, the competency of actual test cases is based on digital experts in Thailand. For example, the competency required by an American blogger might differ from the competency needed by a Thai blogger. This means that the prediction results may need to be more accurate in some cases. Additionally, there may be instances where a student's competencies are equidistant between two or more careers in a multidimensional space. In such cases, the career prediction for these students may require more work.

CONCLUSION

This article introduces CARIA, a career recommendation system that offers personalized career recommendations using similarity measures. The system evaluates the match between a student's skills and each career option to determine the most suitable career choices based on increasing similarity scores. To assess its accuracy, the system's performance was tested using data from 100 different career rankings, and the precision@10 metric was used. The evaluation results demonstrated that CARIA outperforms other career recommendation techniques, achieving a precision@10 score of 0.83. Furthermore, the system's usability was assessed using the SUMI tool, which revealed that the system's usability is generally good, with an average score of 2.66 and a standard deviation of 0.56. This indicates that the career recommendation system is practical and effective.

However, the system has some limitations that need to be addressed. For instance, experts must evaluate new careers to determine their competency before they can be added to the system. Moreover, some careers have the same ranking because they share the same competency score, which may require identifying additional competencies specific to those careers. To enhance the system's performance, future work will explore the use of other techniques, such as fuzzy logic and statistical methods.

COMPETING INTERESTS STATEMENT

The authors of this publication declare there are no competing interests.

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